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```
> #Student Information System
> roll_no <- 417
> name <- "Aravind"
> cgpa <- 8.75
> pass <- TRUE
> roll_no
[1] 417
> name
[1] "Aravind"
> cgpa
[1] 8.75
> pass
[1] TRUE
```

```
> #Employee Salary Record
> emp_id <- 589
> salary <- 850000
> department <- "ECE"
> is_permanent <- FALSE
> emp_id
[1] 589
> salary
[1] 850000
> department
[1] "ECE"
> is_permanent
[1] FALSE
```

```
> #Online Shopping Cart
> product_name <- "Smartphone"
> price <- 45000
> stock_available <- 27
> discount_eligible <- TRUE
> product_name
[1] "Smartphone"
> price
[1] 45000
> stock_available
[1] 27
> discount_eligible
[1] TRUE
```

```
> #Weather Monitoring System
> city <- "Coimbatore"
> temperature <- 28.3
> rain <- TRUE
> wind_speed <- 14.6
> city
[1] "Coimbatore"
> temperature
[1] 28.3
> rain
[1] TRUE
> wind_speed
[1] 14.6
```

```
> #Examination Result Processing
> subject <- "Mathematics"
> total_marks <- 100
> marks_obtained <- 76
> result <- marks_obtained >= 40
> subject
[1] "Mathematics"
> total_marks
[1] 100
> marks_obtained
[1] 76
> result
[1] TRUE
```

```
> #Banking Application
> account_no <- 458921367
> holder_name <- "Karthik"
> balance <- 725000.50
> account_active <- TRUE
> account_no
[1] 458921367
> holder_name
[1] "Karthik"
> balance
[1] 725000.5
> account_active
[1] TRUE
```

```
> #Movie Rating System
> movie_title <- "Vikram"
> rating <- 8.7
> reviews <- 215
> available <- TRUE
> movie_title
[1] "Vikram"
> rating
[1] 8.7
> reviews
[1] 215
> available
[1] TRUE
```

```
> #Traffic Monitoring System
> vehicle_count <- 528
> location <- "Anna Nagar"
> avg_speed <- 42.8
> congested <- FALSE
> vehicle_count
[1] 528
> location
[1] "Anna Nagar"
> avg_speed
[1] 42.8
> congested
[1] FALSE
```

```
> #Loan Eligibility Check
> age <- 23.7
> income <- 31785
> if(age > 21 & income > 30000){
+   print("Eligible for Loan")
+ } else {
+   print("Not Eligible for Loan")
+ }
[1] "Eligible for Loan"
```

```
> #Attendance Requirement
> attendance <- 73.6
> if(attendance >= 75){
+   print("Eligible for Exam")
+ } else {
+   print("Not Eligible for Exam")
+ }
[1] "Not Eligible for Exam"
```

```
> #Online Discount System
> amount <- 5099.5
> premium_member <- FALSE
> if(amount > 5000 | premium_member){
+   print("Discount Applied")
+ } else {
+   print("No Discount")
+ }
[1] "Discount Applied"
```

```
> #Electricity Bill Slab
> units <- 199.3
> if(units < 100){
+   print("Low Usage")
+ } else if(units <= 300){
+   print("Medium Usage")
+ } else {
+   print("High Usage")
+ }
[1] "Medium Usage"
```

```
> #Employee Bonus Eligibility
> experience <- 5.2
> rating <- 4.01
> if(experience > 5 & rating > 4){
+   print("Bonus Eligible")
+ } else {
+   print("Not Eligible for Bonus")
+ }
[1] "Bonus Eligible"
```

```
> #Password Validation
> password <- "z9$K1pLx"
> if(nchar(password) >= 8){
+   print("Valid Password")
+ } else {
+   print("Invalid Password")
+ }
[1] "Valid Password"
```

```
> #Temperature Warning System
> temperature <- -2.4
> if(temperature < 0 | temperature > 40){
+   print("Extreme Temperature Warning")
+ } else {
+   print("Normal Temperature")
+ }
[1] "Extreme Temperature Warning"
```

```
> #Grading System
> marks <- 74.8
> if(marks >= 90){
+   grade <- "A"
+ } else if(marks >= 75){
+   grade <- "B"
+ } else if(marks >= 50){
+   grade <- "C"
+ } else {
+   grade <- "Fail"
+ }
> print(grade)
[1] "C"
```

```
> #Bank Credit Card Approval System
> age <- 21.5
> income <- 25001
> credit_score <- 699.5
> if(age >= 21 & age <= 60 & income >= 25000 & credit_score >= 700){
+   print("Credit Card Approved")
+ } else {
+   print("Credit Card Rejected")
+ }
[1] "Credit Card Rejected"
> 2(j)
```

```
> bp <- 139.7
> sugar <- 201.3
> if(bp > 140 & sugar > 200){
+   risk <- "High Risk"
+ } else if((bp >= 120 & bp <= 140) | (sugar >= 140 & sugar
+   risk <- "Medium Risk"
+ } else {
+   risk <- "Low Risk"
+ }
> print(risk)
[1] "Medium Risk"
```

```
> vehicle_count <- 101.9
> if(vehicle_count > 200){
+   print("Long Green Signal")
+ } else if(vehicle_count >= 100){
+   print("Medium Green Signal")
+ } else {
+   print("Short Green Signal")
+ }
[1] "Medium Green Signal"
```

```
> #Bank Credit Card Approval System
> age <- 21.5
> income <- 25001
> credit_score <- 699.5
> if(age >= 21 & age <= 60 & income >= 25000 & credit_score >= 700){
+   print("Credit Card Approved")
+ } else {
+   print("Credit Card Rejected")
+ }
[1] "Credit Card Rejected"
> 2(j)
```

```
> #Employee Performance Appraisal System
> attendance <- 89.9
> performance <- 7.99
> project_completed <- TRUE
> if(attendance >= 90 & performance >= 8 & project_completed){
+   print("Eligible for Appraisal")
+ } else {
+   print("Not Eligible for Appraisal")
+ }
[1] "Not Eligible for Appraisal"
```

```
> face_detected <- TRUE
> multiple_faces <- TRUE
> internet_connected <- TRUE
> if(!face_detected | multiple_faces | !internet_connected){
+   print("Exam Disqualified")
+ } else {
+   print("Exam Allowed")
+ }
[1] "Exam Disqualified"
```

```
> #Dynamic Pricing in Ride-Hailing App
> high_demand <- FALSE
> available_drivers <- 49.5
> if(high_demand & available_drivers < 50){
+   print("Surge Pricing Applied")
+ } else {
+   print("Normal Pricing")
+ }
[1] "Normal Pricing"
```

```
> #Industrial Machine Safety System
> temperature <- 79.9
> pressure <- 121.2
> vibration <- 6.9
> if(temperature > 80 | pressure > 120 | vibration > 7){
+   print("Machine Shutdown")
+ } else {
+   print("Machine Running Normally")
+ }
[1] "Machine Shutdown"
```

```
> #Bank Interest Calculator
> simple_interest <- function(p, r, t) {
+   si <- (p * r * t) / 100
+   return(si)
+ }
> simple_interest(12000, 6, 3)
[1] 2160
> 3(b)
```

```
> #Student Result Processing
> check_result <- function(marks){
+   if(marks >= 30){
+     return("Pass")
+   } else {
+     return("Fail")
+   }
+ }
> check_result(45)
[1] "Pass"
> 3(c)
```

```
> #Employee Salary Calculation
> net_salary<- function(
+     basic_salary
+ ){
+   tax <-basic_salary* 0.10
+   net <-basic_salary- tax
+   return(net)
+ }
> net_salary
function(
    basic_salary
){
  tax <-basic_salary* 0.10
  net <-basic_salary- tax
  return(net)
}
> (28000)
[1] 28000
> 3(d)

> #Electricity Bill Generator
> final_amount <- function(amount){
+   if(amount >= 25000){
+     discount <- amount * 0.20
+   } else if(amount >= 2000){
+     discount <- amount * 0.10
+   } else {
+     discount <- 0
+   }
+   return(amount - discount)
+ }
> final_amount(18000)
[1] 16200
> 3(e)
```

```
> #Electricity Bill Generator
> final_amount <- function(amount){
+   if(amount >= 25000){
+     discount <- amount * 0.20
+   } else if(amount >= 2000){
+     discount <- amount * 0.10
+   } else {
+     discount <- 0
+   }
+   return(amount - discount)
+ }
> final_amount(18000)
[1] 16200
> 3(e)
```

```
> #Temperature Conversion System
> celsius_to_fahrenheit<- function(c){
+   f <- (c * 9/5) + 32
+   return(f)
+ }
> celsius_to_fahrenheit
function(c){
  f <- (c * 9/5) + 32
  return(f)
}
> (30)
[1] 30
> 3(f)
```

```
> #Grading System
> assign_grade<- function(marks){
+   if(marks >= 90){
+     "A"
+   } else if(marks >= 75){
+     "B"
+   } else if(marks >= 50){
+     "C"
+   } else {
+     "Fail"
+   }
+ }
> assign_grade
function(marks){
  if(marks >= 90){
    "A"
  } else if(marks >= 75){
    "B"
  } else if(marks >= 50){
    "C"
  } else {
    "Fail"
  }
}
> (82)
[1] 82
> 3(g)
```

```
> #Voting Eligibility Checker
> check_voting<- function(age){
+   if(age >= 18){
+     return("Eligible to Vote")
+   } else {
+     return("Not Eligible to Vote")
+   }
+ }
> check_voting
function(age){
  if(age >= 18){
    return("Eligible to Vote")
  } else {
    return("Not Eligible to Vote")
  }
}
> (22)
[1] 22
> 3(h)
```

```
> #Bank Loan EMI Calculator System
> calculate_emi<- function(principal, rate, years){
+   monthly_rate<- rate / (12 * 100)
+   months <- years * 12
+   emi<- (principal *monthly_rate* (1 +monthly_rate)^months) /((1 +monthly_rate)^months - 1)
+   return(round(emi, 2))
+ }
> calculate_emi(400000,8,5)
[1] 8110.56
> 3(i)
```

```
> #Hospital Patient Billing System
> hospital_bill<- function(consultation,room_charge, days){
+   subtotal <- consultation + (room_charge* days)
+   tax <- subtotal * 0.05
+   total <- subtotal + tax
+   return(total)
+ }
> hospital_bill(3000,15000,10)
[1] 160650
> 3(j)
```

```
function(
    cart_value
)
{
  if(cart_value>= 3000){
    discount <-cart_value* 0.10
  } else {
    discount <- 0
  }
  if(
    cart_value>= 5000){
    delivery <- 0
  } else {
    delivery <- 100
  }
  final_amount<-cart_value- discount + delivery
  return
  (
    final_amount
  )
}
> (3500)
[1] 3500
> 3(k)
```

```
> #University GPA Calculation System
> calculate_gpa<- function(marks){
+   avg<- mean(marks)
+   if(avg>= 85){
+     result <- "Distinction"
+   } else if(avg>= 70){
+     result <- "First Class"
+   } else if(avg>= 50){
+     result <- "Second Class"
+   } else {
+     result <- "Fail"
+   }
+   return(result)
+ }
> calculate_gpa
function(marks){
  avg<- mean(marks)
  if(avg>= 85){
    result <- "Distinction"
  } else if(avg>= 70){
    result <- "First Class"
  } else if(avg>= 50){
    result <- "Second Class"
  } else {
    result <- "Fail"
  }
  return(result)
}
> (c(70, 65, 80, 75, 85))
[1] 70 65 80 75 85
> 3(l)
```

```
> #Smart Electricity Meter System
> smart_meter_bill<- function(peak_units,off_peak_units){
+   bill <- (peak_units* 6) + (off_peak_units* 3)
+   total_units<-peak_units+off_peak_units
+   if(total_units> 300){
+     bill <- bill * 1.10
+   }
+   return(bill)
+ }
> smart_meter_bill(200,100)
[1] 1500
> 3(m)
```

```
> #Insurance Premium Calculation System
> insurance_premium<- function(base, age, smoker){
+   premium <- base
+   if(age > 50){
+     premium <- premium * 1.20
+   }
+   if(smoker){
+     premium <- premium * 1.30
+   }
+   return(premium)
+ }
> insurance_premium(300000, 45, FALSE)
[1] 3e+05
> 3(n)
```

```
> #Employee Payroll Processing System
> payroll <- function(basic){
+   hra<- basic * 0.20
+   da <- basic * 0.10
+   gross <- basic +hra+ da
+   tax <- gross * 0.12
+   net <- gross - tax
+   return(net)
+ }
> payroll(35000)
[1] 40040
> 3(o)
```

```
> #Smart City Water Usage Monitoring
> water_usage_status<- function(litres){
+   if(litres <= 50){
+     "Normal"
+   } else if(litres <= 100){
+     "High Usage"
+   } else {
+     "Critical"
+   }
+ }
> water_usage_status
function(litres){
  if(litres <= 50){
    "Normal"
  } else if(litres <= 100){
    "High Usage"
  } else {
    "Critical"
  }
}
> (80)
[1] 80
```

```
> students <- data.frame(
+   RollNo = c(482, 483, 484),
+   Name = c("Kiran", "Megha", "Sanjay"),
+   Dept = c("IT", "CSE", "EEE")
+ )
> marks <- data.frame(
+   Marks = c(78, 85, 92)
+ )
> student_data <- cbind(students, marks)
> student_data
  RollNo   Name Dept Marks
1    482  Kiran   IT    78
2    483  Megha  CSE    85
3    484 Sanjay  EEE    92
> new_students <- data.frame(
+   RollNo = c(485),
+   Name = c("Divya"),
+   Dept = c("ECE"),
+   Marks = c(81)
+ )
> final_data <- rbind(student_data, new_students)
> final_data
  RollNo   Name Dept Marks
1    482  Kiran   IT    78
2    483  Megha  CSE    85
3    484 Sanjay  EEE    92
4    485  Divya  ECE    81
> 4.2: Employee Management System
```

```
> employee_details <- data.frame(
+   EmpID = c(11, 12, 13),
+   Name = c("ARUN", "KAVYA", "MOHAN"),
+   Dept = c("Finance", "HR", "IT")
+ )
> salary_details <- data.frame(
+   Salary = c(30000, 45000, 60000)
+ )
> employees <- cbind(employee_details, salary_details)
> employees
  EmpID  Name    Dept Salary
1    11  ARUN Finance  30000
2    12 KAVYA      HR  45000
3    13 MOHAN      IT  60000
> new_employee <- data.frame(
+   EmpID = 14,
+   Name = "Priya",
+   Dept = "HR",
+   Salary = 48000
+ )
> employees <- rbind(employees, new_employee)
> employees
  EmpID  Name    Dept Salary
1    11  ARUN Finance  30000
2    12 KAVYA      HR  45000
3    13 MOHAN      IT  60000
4    14 Priya      HR  48000
> 4.3: Hospital Patient Records
```

```
> patient_info <- data.frame(
+   PatientID = c(410, 411),
+   Name = c("Ramesh", "Geetha"),
+   Age = c(42, 36)
+ )
> treatment_cost <- data.frame(
+   Cost = c(18000, 22000)
+ )
> patient_records <- cbind(patient_info, treatment_cost)
> patient_records
  PatientID   Name Age  Cost
1       410 Ramesh  42 18000
2       411 Geetha  36 22000
> new_patient <- data.frame(
+   PatientID = 412,
+   Name = "Vijay",
+   Age = 50,
+   Cost = 27000
+ )
> patient_records <- rbind(patient_records, new_patient)
> patient_records
  PatientID   Name Age  Cost
1       410 Ramesh  42 18000
2       411 Geetha  36 22000
3       412  Vijay  50 27000
> 4.4: Product Inventory System
```

```
> products <- data.frame(
+   ProductID = c(10, 20, 30),
+   ProductName = c("Notebook", "Eraser", "Marker")
+ )
> stock <- data.frame(
+   Quantity = c(200, 300, 150)
+ )
> inventory <- cbind(products, stock)
> inventory
  ProductID ProductName Quantity
1        10    Notebook      200
2        20      Eraser      300
3        30      Marker      150
> new_product <- data.frame(
+   ProductID = 40,
+   ProductName = "Scale",
+   Quantity = 180
+ )
> inventory <- rbind(inventory, new_product)
> inventory
  ProductID ProductName Quantity
1        10    Notebook      200
2        20      Eraser      300
3        30      Marker      150
4        40       Scale      180
> 4.5: University Academic Management System
```

```
> students <- data.frame(
+   RollNo = c(21, 22, 23),
+   Name = c("Anu", "Teja", "Varun"),
+   Dept = c("IT", "CSE", "ECE")
+ )
> marks <- data.frame(
+   Sem1 = c(75, 88, 91),
+   Sem2 = c(80, 84, 93)
+ )
> attendance <- data.frame(
+   Attendance = c(89, 91, 94)
+ )
> student_records <- cbind(students, marks, attendance)
> student_records
  RollNo  Name Dept Sem1 Sem2 Attendance
1     21   Anu   IT   75   80         89
2     22  Teja  CSE   88   84         91
3     23 Varun  ECE   91   93         94
> new_students <- data.frame(
+   RollNo = c(24, 25),
+   Name = c("Nisha", "Ravi"),
+   Dept = c("EEE", "IT"),
+   Sem1 = c(82, 86),
+   Sem2 = c(85, 89),
+   Attendance = c(90, 92)
+ )
> final_records <- rbind(student_records, new_students)
> final_records
  RollNo  Name Dept Sem1 Sem2 Attendance
1     21   Anu   IT   75   80         89
2     22  Teja  CSE   88   84         91
3     23 Varun  ECE   91   93         94
4     24 Nisha  EEE   82   85         90
5     25  Ravi   IT   86   89         92
> 4.6: Corporate Employee Payroll System
```

```
> emp_details <- data.frame(
+   EmpID = c(401, 402),
+   Name = c("SURESH", "LATHA"),
+   Dept = c("HR", "Finance")
+ )
> salary <- data.frame(
+   Basic = c(28000, 36000),
+   HRA = c(5000, 7500),
+   DA = c(2500, 3200)
+ )
> rating <- data.frame(
+   Rating = c(4.1, 4.6)
+ )
> payroll <- cbind(emp_details, salary, rating)
> payroll
  EmpID   Name    Dept Basic  HRA   DA Rating
1   401 SURESH      HR 28000 5000 2500    4.1
2   402  LATHA Finance 36000 7500 3200    4.6
> new_emp <- data.frame(
+   EmpID = 403,
+   Name = "KIRAN",
+   Dept = "IT",
+   Basic = 42000,
+   HRA = 8500,
+   DA = 4200,
+   Rating = 4.7
+ )
> payroll <- rbind(payroll, new_emp)
> payroll
  EmpID   Name    Dept Basic  HRA   DA Rating
1   401 SURESH      HR 28000 5000 2500    4.1
2   402  LATHA Finance 36000 7500 3200    4.6
3   403  KIRAN      IT 42000 8500 4200    4.7
> 4.7: Hospital Information System
```

```
> patients <- data.frame(
+   PatientID = c(501, 502),
+   Name = c("RAJU", "MANI"),
+   Age = c(47, 39)
+ )
> diagnosis <- data.frame(
+   Disease = c("Asthma", "Fever")
+ )
> billing <- data.frame(
+   Amount = c(14000, 11000)
+ )
> hospital_data <- cbind(patients, diagnosis, billing)
> hospital_data
  PatientID Name Age Disease Amount
1       501 RAJU  47  Asthma  14000
2       502 MANI  39   Fever  11000
> new_patient <- data.frame(
+   PatientID = 503,
+   Name = "KUMAR",
+   Age = 55,
+   Disease = "Cancer",
+   Amount = 30000
+ )
> hospital_data <- rbind(hospital_data, new_patient)
> hospital_data
  PatientID  Name Age Disease Amount
1       501  RAJU  47  Asthma  14000
2       502  MANI  39   Fever  11000
3       503 KUMAR  55  Cancer  30000
> 4.8: Retail Sales Analytics System
```

```
> products <- data.frame(
+   ProductID = c(31, 32, 33),
+   ProductName = c("TV", "Fridge", "Washing Machine")
+ )
> sales_qty <- data.frame(
+   UnitsSold = c(70, 50, 40)
+ )
> revenue <- data.frame(
+   Revenue = c(1400000, 2500000, 1600000)
+ )
> sales_data <- cbind(products, sales_qty, revenue)
> sales_data
  ProductID     ProductName UnitsSold Revenue
1        31              TV        70 1400000
2        32          Fridge        50 2500000
3        33 Washing Machine        40 1600000
> new_product <- data.frame(
+   ProductID = 34,
+   ProductName = "AC",
+   UnitsSold = 60,
+   Revenue = 1800000
+ )
> sales_data <- rbind(sales_data, new_product)
> sales_data
  ProductID     ProductName UnitsSold Revenue
1        31              TV        70 1400000
2        32          Fridge        50 2500000
3        33 Washing Machine        40 1600000
4        34              AC        60 1800000
> 4.9: Smart Agriculture Monitoring System
```

```
> field_info <- data.frame(
+   FieldID = c(5, 6),
+   Crop = c("Corn", "Barley")
+ )
> sensor_data <- data.frame(
+   Moisture = c(50, 45),
+   Temp = c(30, 28)
+ )
> yield <- data.frame(
+   ExpectedYield = c(3.9, 4.2)
+ )
> agri_data <- cbind(field_info, sensor_data, yield)
> agri_data
  FieldID   Crop Moisture Temp ExpectedYield
1       5   Corn       50   30           3.9
2       6 Barley       45   28           4.2
> new_field <- data.frame(
+   FieldID = 7,
+   Crop = "Soybean",
+   Moisture = 42,
+   Temp = 29,
+   ExpectedYield = 4.0
+ )
> agri_data <- rbind(agri_data, new_field)
> agri_data
  FieldID    Crop Moisture Temp ExpectedYield
1       5    Corn       50   30           3.9
2       6  Barley       45   28           4.2
3       7 Soybean       42   29           4.0
> 4.10: University Academic Management System
```

```
> students <- data.frame(
+   RollNo = c(40, 50, 60),
+   Name = c("KAVYA", "ARUN", "MEENA"),
+   Dept = c("IT", "CSE", "ECE")
+ )
> marks <- data.frame(
+   RollNo = c(40, 50, 60),
+   Sem1 = c(81, 76, 89),
+   Sem2 = c(83, 79, 91)
+ )
> attendance <- data.frame(
+   RollNo = c(40, 50, 60),
+   Attendance = c(88, 92, 90)
+ )
> student_data <- merge(students, marks, by = "RollNo")
> final_records <- merge(student_data, attendance, by = "RollNo")
> final_records
  RollNo  Name Dept Sem1 Sem2 Attendance
1     40 KAVYA   IT   81   83         88
2     50  ARUN  CSE   76   79         92
3     60 MEENA  ECE   89   91         90
> 4.11: Corporate Employee Payroll System
```

```
> emp_details <- data.frame(
+   EmpID = c(201, 202, 203),
+   Name = c("VIJAY", "ANITA", "RAHUL"),
+   Dept = c("Finance", "IT", "HR")
+ )
> salary <- data.frame(
+   EmpID = c(201, 202),
+   Basic = c(32000, 37000),
+   HRA = c(6500, 7200),
+   DA = c(3100, 3600)
+ )
> rating <- data.frame(
+   EmpID = c(201, 203),
+   Rating = c(4.2, 4.9)
+ )
> payroll <- merge(emp_details, salary, by = "EmpID", all.x = TRUE)
> payroll <- merge(payroll, rating, by = "EmpID", all.x = TRUE)
> payroll
  EmpID  Name    Dept Basic  HRA   DA Rating
1   201 VIJAY Finance 32000 6500 3100    4.2
2   202 ANITA      IT 37000 7200 3600     NA
3   203 RAHUL      HR    NA   NA   NA    4.9
> 4.12: Hospital Information System
```

```
> patients <- data.frame(
+   PatientID = c(301, 302, 303),
+   Name = c("RAMESH", "PRIYA", "KIRAN"),
+   Age = c(40, 29, 33)
+ )
> diagnosis <- data.frame(
+   PatientID = c(301, 303),
+   Disease = c("Asthma", "Heart Disease")
+ )
> billing <- data.frame(
+   PatientID = c(301, 302, 303),
+   Amount = c(18000, 15000, 27000)
+ )
> hospital_data <- merge(patients, diagnosis, by = "PatientID", all.x = TRUE)
> hospital_data <- merge(hospital_data, billing, by = "PatientID")
> hospital_data
  PatientID   Name Age       Disease Amount
1       301 RAMESH  40        Asthma  18000
2       302  PRIYA  29          <NA>  15000
3       303  KIRAN  33 Heart Disease  27000
> 4.13: Retail Sales Analytics System
```

```
> products <- data.frame(
+   ProductID = c(51, 52, 53, 54),
+   ProductName = c("TV", "Refrigerator", "Microwave", "Smartwatch")
+ )
> sales_qty <- data.frame(
+   ProductID = c(51, 52, 53),
+   UnitsSold = c(65, 45, 30)
+ )
> revenue <- data.frame(
+   ProductID = c(51, 52, 54),
+   Revenue = c(1500000, 2200000, 950000)
+ )
> sales_data <- merge(products, sales_qty, by = "ProductID", all = TRUE)
> sales_data <- merge(sales_data, revenue, by = "ProductID", all = TRUE)
> sales_data
  ProductID  ProductName UnitsSold Revenue
1        51           TV        65 1500000
2        52 Refrigerator        45 2200000
3        53    Microwave        30      NA
4        54   Smartwatch        NA  950000
> 4.14: Smart Agriculture Monitoring System
```

```
> field_info <- data.frame(
+   FieldID = c(10, 20, 30),
+   Crop = c("Cotton", "Sugarcane", "Groundnut")
+ )
> sensor_data <- data.frame(
+   FieldID = c(10, 20),
+   Moisture = c(48, 52),
+   Temp = c(31, 30)
+ )
> yield <- data.frame(
+   FieldID = c(10, 30),
+   ExpectedYield = c(5.2, 4.9)
+ )
> agri_data <- merge(field_info, sensor_data, by = "FieldID", all.x = TRUE)
> agri_data <- merge(agri_data, yield, by = "FieldID", all.x = TRUE)
> agri_data
  FieldID      Crop Moisture Temp ExpectedYield
1      10    Cotton       48   31           5.2
2      20 Sugarcane       52   30            NA
3      30 Groundnut       NA   NA           4.9
```

```
> names <- c("KIRAN", "DEEPA", "VARUN")
> upper_names <- toupper(names)
> name_length <- nchar(upper_names)
> upper_names
[1] "KIRAN" "DEEPA" "VARUN"
> name_length
[1] 5 5 5
> 5.2: Student Email Processing
```

```
> emails <- c("MEENA@gmail.com", "ARUN@yahoo.com", "KAVYA@outlook.com")
> usernames <- sub("@.*", "", emails)
> usernames
[1] "MEENA" "ARUN"  "KAVYA"
> 5.3: Online Form Input Cleanup
```

```
> input <- c(" Machine Learning ", " Cloud Computing ", " Big Data ")
> clean_input <- trimws(input)
> clean_input
[1] "Machine Learning" "Cloud Computing"  "Big Data"
> 5.4: Product Code Validation
```

```
> product_codes <- c("PRD601", "PRD602", "XYZ603")
> valid_codes <- startsWith(product_codes, "PRD")
> valid_codes
[1]  TRUE  TRUE FALSE
> 5.5: Text Replacement in Reports
```

```
> report <- "warning in module, warning in code, warning detected"
> updated_report <- gsub("warning", "notice", report)
> updated_report
[1] "notice in module, notice in code, notice detected"
> 5.6: Password Strength Checker
```

```
> password <- "KAVYA2024"
> if(nchar(password) >= 8){
+   print("Strong Password")
+ } else {
+   print("Weak Password")
+ }
[1] "Strong Password"
> 5.7: Sentence Word Count
```

```
> sentence <- "Python is widely used in data analysis"
> word_count <- length(strsplit(sentence, " ")[[1]])
> word_count
[1] 7
> 5.8: File Extension Extraction
```

```
> files <- c("notes.txt", "presentation.pptx", "photo.jpg")
> extensions <- sub(".*\\.", "", files)
> extensions
[1] "txt"  "pptx" "jpg"
> 5.9: Student Name Abbreviation
```

```
> names <- c("PRAVEEN", "SANGEETHA", "HARISH")
> abbr <- substr(names, 1, 3)
> abbr
[1] "PRA" "SAN" "HAR"
> 5.10: Search Operation in Text
```

```
> names <- c("KARTHIK", "ANITA", "SURESH", "DIVYA")
> matched_names <- grep("a", names, value = TRUE)
> matched_names
character(0)
> 5.11: University Student Data Cleaning System
```

```
> students <- data.frame(
+   Name = c(" Kavya ", "ARUN ", " meera"),
+   Email = c("Kavya@UNIV.edu", "arun@univ.edu", "meera@gmail.com"),
+   Department = c(" it", "CSE ", " Ece ")
+ )
> students
     Name           Email Department
1  Kavya   Kavya@UNIV.edu         it
2   ARUN     arun@univ.edu        CSE
3   meera meera@gmail.com        Ece
> 5.12: Product Inventory Text Cleaning
```

```
> products <- data.frame(
+   Product = c(" oneplus phone ", "LG-TV ", " hp laptop"),
+   Code = c("PRD-401-ONE", "PRD-402-LG", "PRD-403-HP")
+ )
> products$Product <- trimws(products$Product)
> products$Product <- tools::toTitleCase(tolower(products$Product))
> products$ProductCode <- sub("PRD-[0-9]+-", "", products$Code)
> products
        Product        Code ProductCode
1 Oneplus Phone PRD-401-ONE         ONE
2         Lg-Tv  PRD-402-LG          LG
3     Hp Laptop  PRD-403-HP          HP
> 5.13: Social Media Comment Moderation System
```

```
> comments <- data.frame(
+   User = c("X", "Y", "Z"),
+   Comment = c(
+     "This service is poor",
+     "Really POOR quality",
+     "Not poor at all"
+   )
+ )
> comments$Comment <- tolower(comments$Comment)
> comments$CleanComment <- gsub("poor", "unsatisfactory", comments$Comment)
> comments
  User              Comment                   CleanComment
1    X this service is poor this service is unsatisfactory
2    Y  really poor quality  really unsatisfactory quality
3    Z      not poor at all      not unsatisfactory at all
>
```

```
> marks <- c(78, 84, 91, 67, 73)
> total <- sum(marks)
> average <- mean(marks)
> highest <- max(marks)
> total
[1] 393
> average
[1] 78.6
> highest
[1] 91
> 6.2: Temperature Monitoring System (Vectors)
```

```
> temperature <- c(25, 27, 31, 29, 33, 28, 30)
> min_temp <- min(temperature)
> max_temp <- max(temperature)
> min_temp
[1] 25
> max_temp
[1] 33
> 6.3: Employee Profile Storage (Lists)
```

```
> employee <- list(
+   ID = 401,
+   Name = "Karthik",
+   Salary = 68000,
+   Department = "HR"
+ )
> employee
$ID
[1] 401

$Name
[1] "Karthik"

$Salary
[1] 68000

$Department
[1] "HR"

> employee$Name
[1] "Karthik"
> employee$Salary
[1] 68000
> 6.4: Patient Medical Record System (Lists)
```

```
> patient <- list(
+   PatientID = 602,
+   Name = "Suresh",
+   Age = 42,
+   TestResults = c(110, 135, 128)
+ )
> patient
$PatientID
[1] 602

$Name
[1] "Suresh"

$Age
[1] 42

$TestResults
[1] 110 135 128

> patient$TestResults
[1] 110 135 128
> 6.5: College Student Database (Data Frames)
```

```
> students <- data.frame(
+   RollNo = c(31, 32, 33),
+   Name = c("RAHUL", "SNEHA", "DIVYA"),
+   Dept = c("IT", "CSE", "ECE"),
+   Marks = c(88, 76, 90)
+ )
> students
  RollNo  Name Dept Marks
1     31 RAHUL   IT    88
2     32 SNEHA  CSE    76
3     33 DIVYA  ECE    90
> 6.6: Sales Report System (Data Frames)
```

```
> sales <- data.frame(
+   Month = c("Apr", "May", "Jun"),
+   Sales = c(150000, 720000, 880000)
+ )
> total_sales <- sum(sales$Sales)
> total_sales
[1] 1750000
> 6.7: E-Commerce Order System (Combination of Data Structures)
```

```
> prices <- c(1800, 1200, 650)
> customer <- list(
+   CustomerID = 402,
+   Name = "MEERA",
+   City = "Bangalore"
+ )
> orders <- data.frame(
+   Item = c("Tablet", "Keyboard", "Mouse"),
+   Price = prices
+ )
> prices
[1] 1800 1200  650
> customer
$CustomerID
[1] 402

$Name
[1] "MEERA"

$City
[1] "Bangalore"

> orders
      Item Price
1   Tablet  1800
2 Keyboard  1200
3    Mouse   650
> 6.8: Online Exam System
```

```
> subject_marks <- c(82, 91, 85)
> student_profile <- list(
+   Name = "Kiran",
+   RollNo = 205,
+   Marks = subject_marks
+ )
> class_results <- data.frame(
+   RollNo = c(401, 402, 403),
+   Total = c(250, 265, 275)
+ )
> student_profile
$Name
[1] "Kiran"

$RollNo
[1] 205

$Marks
[1] 82 91 85

> class_results
  RollNo Total
1    401   250
2    402   265
3    403   275
> 6.9: Bank Account Management
```

```
> balances <- c(60000, 58000, 62000)
> customer <- list(
+   AccountNo = 84521,
+   Name = "RAVI"
+ )
> transactions <- data.frame(
+   Date = c("05-01", "06-01", "07-01"),
+   Amount = c(-800, 1500, -300)
+ )
> balances
[1] 60000 58000 62000
> customer
$AccountNo
[1] 84521

$Name
[1] "RAVI"

> transactions
   Date Amount
1 05-01   -800
2 06-01   1500
3 07-01   -300
> 6.10: Real-World Summary Question
```

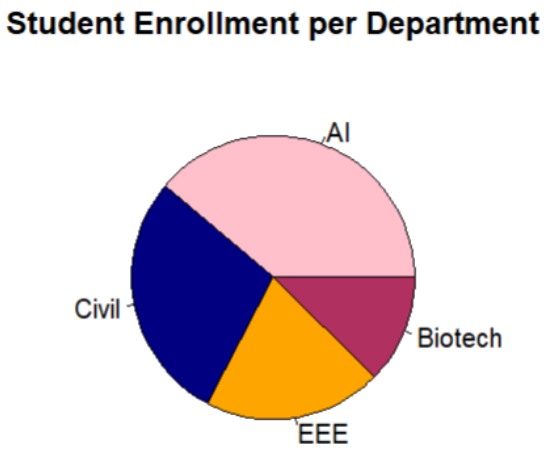
```
> scores <- c(88, 92, 95)
> profile <- list(
+   Name = "KARTHIK",
+   Age = 22,
+   Scores = scores
+ )
> records <- data.frame(
+   Subject = c("English", "Math", "Science"),
+   Marks = scores
+ )
> scores
[1] 88 92 95
> profile
$Name
[1] "KARTHIK"

$Age
[1] 22

$Scores
[1] 88 92 95

> records
  Subject Marks
1 English    88
2    Math    92
3 Science    95
```

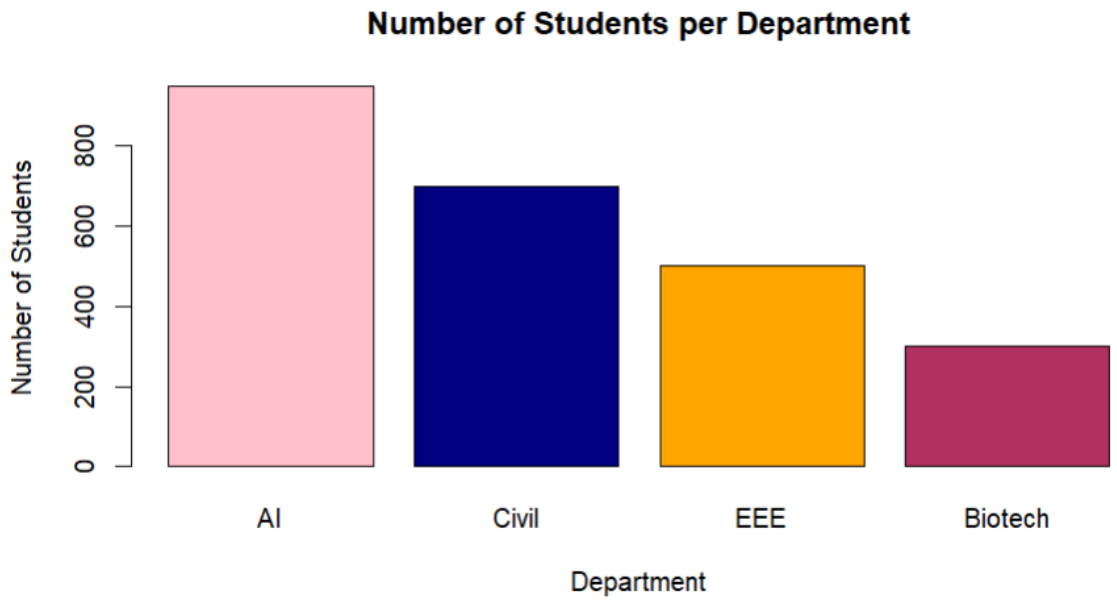
```
> # View first few rows
> head(sales)
        Date Product    Category
1 2026-01-01   Phone Electronics
2 2026-01-01     Bag Accessories
3 2026-01-02  Laptop Electronics
4 2026-01-02 Headset Electronics
5 2026-01-03     Bag Accessories
6 2026-01-03   Phone Electronics
  Quantity Price
1        5 20000
2       10  1500
3        3 50000
4        7  2500
5        5  1500
6        3 20000
> # Check structure and summary
> str(sales)
'data.frame':   6 obs. of  5 variables:
 $ Date    : chr  "2026-01-01" "2026-01-01" "2026-01-02" "2026-01-02" ...
 $ Product : chr  "Phone" "Bag" "Laptop" "Headset" ...
 $ Category: chr  "Electronics" "Accessories" "Electronics" "Electronics"
...
 $ Quantity: int  5 10 3 7 5 3
 $ Price   : int  20000 1500 50000 2500 1500 20000
> summary(sales)
     Date
 Length:6
 Class :character
 Mode  :character
```

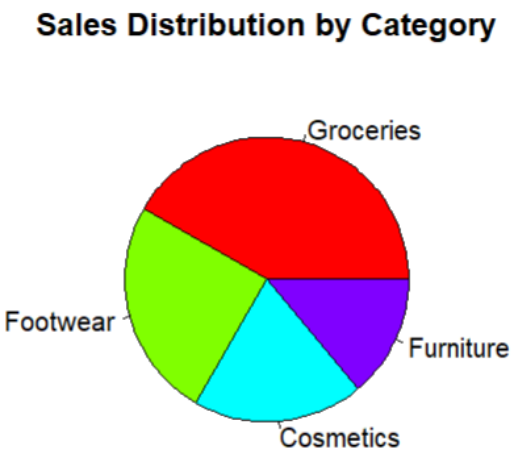
```
   Product
 Length:6
 Class :character
 Mode  :character


   Category
 Length:6
 Class :character
 Mode  :character


    Quantity        Price
 Min.   : 3.0   Min.   : 1500
 1st Qu.: 3.5   1st Qu.: 1750
 Median : 5.0   Median :11250
 Mean   : 5.5   Mean   :15917
 3rd Qu.: 6.5   3rd Qu.:20000
 Max.   :10.0   Max.   :50000
> # Remove extra spaces in Product and Category columns
> sales$Product <- trimws(sales$Product)
> sales$Category <- trimws(sales$Category)
> # Convert Date column to Date format
> sales$Date <- as.Date(sales$Date, format="%Y-%m-%d")
> # Add a column for total sales per row
> sales$TotalSales <- sales$Quantity * sales$Price
> # View cleaned data
> head(sales)
```

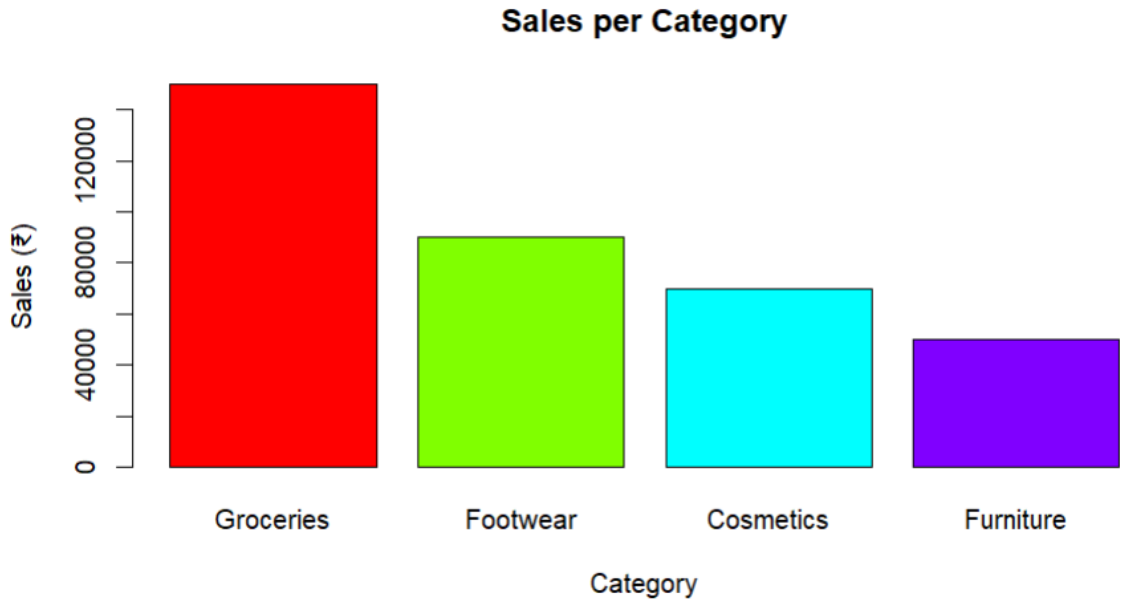
```
> head(sales)
        Date Product    Category
1 2026-01-01   Phone Electronics
2 2026-01-01     Bag Accessories
3 2026-01-02  Laptop Electronics
4 2026-01-02 Headset Electronics
5 2026-01-03     Bag Accessories
6 2026-01-03   Phone Electronics
  Quantity Price TotalSales
1        5 20000     100000
2       10  1500      15000
3        3 50000     150000
4        7  2500      17500
5        5  1500       7500
6        3 20000      60000
> # Calculate total sales for each product
> total_sales_product <- aggregate(TotalSales ~ Product, data=sales, sum)
>
> total_sales_product
  Product TotalSales
1     Bag      22500
2 Headset      17500
3  Laptop     150000
4   Phone     160000
```

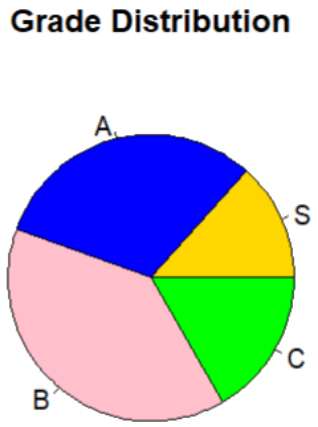
```
> # Calculate total sales for each product
> total_sales_product <- aggregate(TotalSales ~ Product, data=sales, sum)
>
> total_sales_product
  Product TotalSales
1     Bag      22500
2 Headset      17500
3  Laptop     150000
4   Phone     160000
> top_product <- total_sales_product[which.max(total_sales_product$TotalS
ales), ]
> top_product
  Product TotalSales
4   Phone     160000
> quantity_per_category <- aggregate(Quantity ~ Category, data=sales, su
m)
> quantity_per_category
     Category Quantity
1 Accessories       15
2 Electronics       18
> # Products with total sales greater than 10000
> high_sales <- total_sales_product[total_sales_product$TotalSales > 1000
0, ]
> high_sales
  Product TotalSales
1     Bag      22500
2 Headset      17500
3  Laptop     150000
4   Phone     160000
```

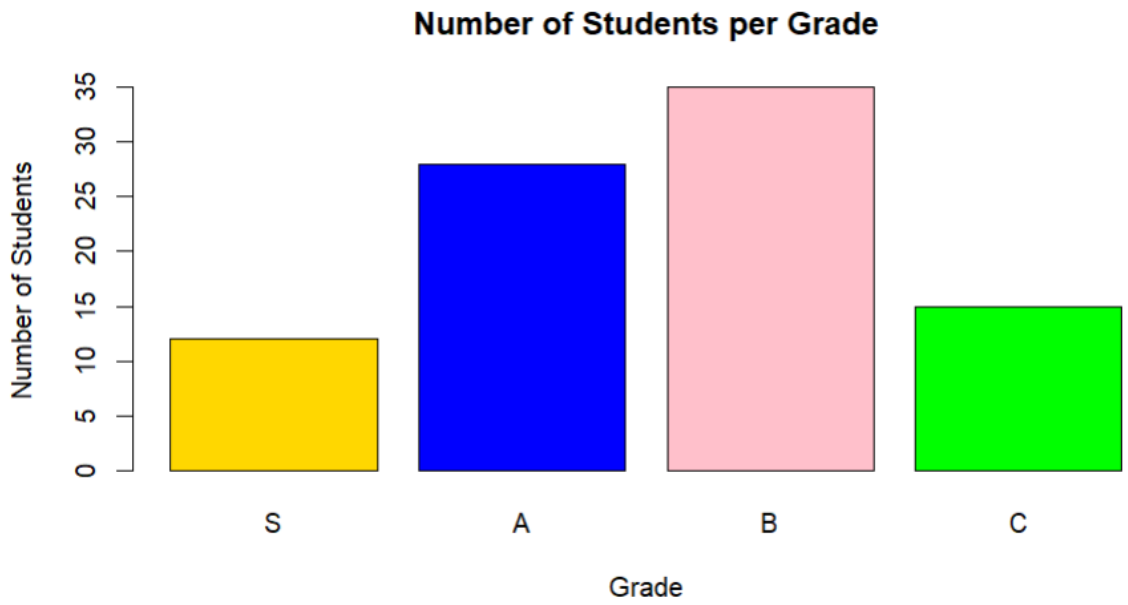

Student Enrollment per Department
AI
Civil
Biotech
EEE

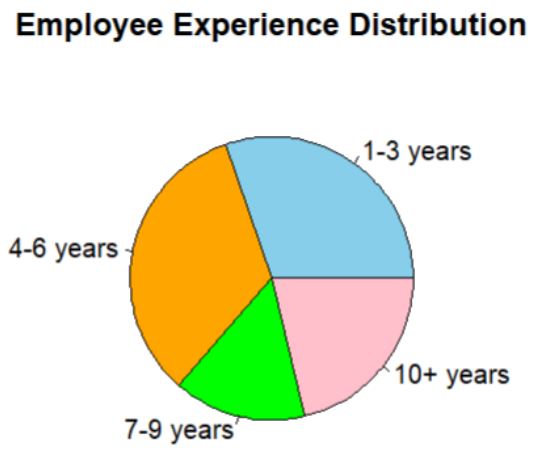

Number of Students per Department
Number of Students
0
200
400
600
800
AI
Civil
EEE
Biotech
Department

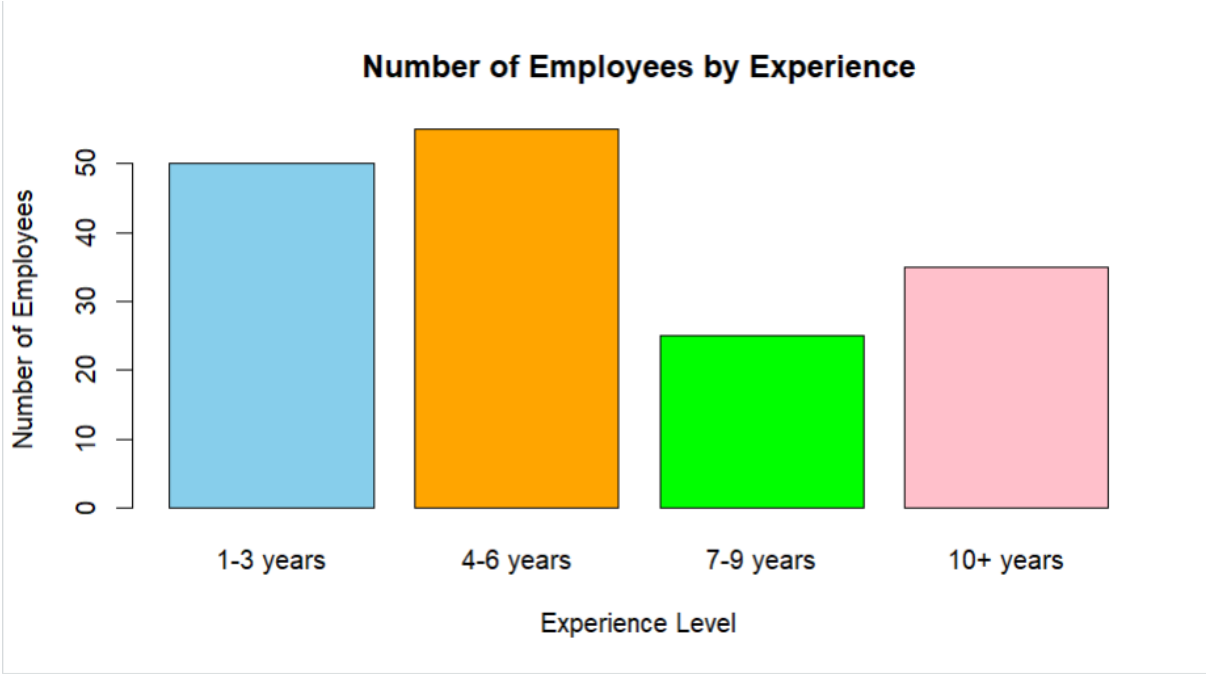

Sales Distribution by Category
Groceries
Footwear
Furniture
Cosmetics

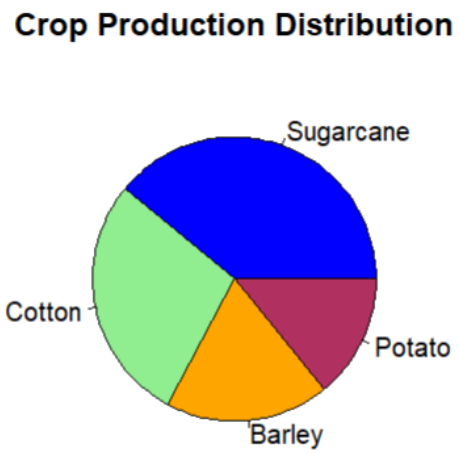

Sales per Category
Sales (₹)
0
40000
80000
120000
Groceries
Footwear
Cosmetics
Furniture
Category

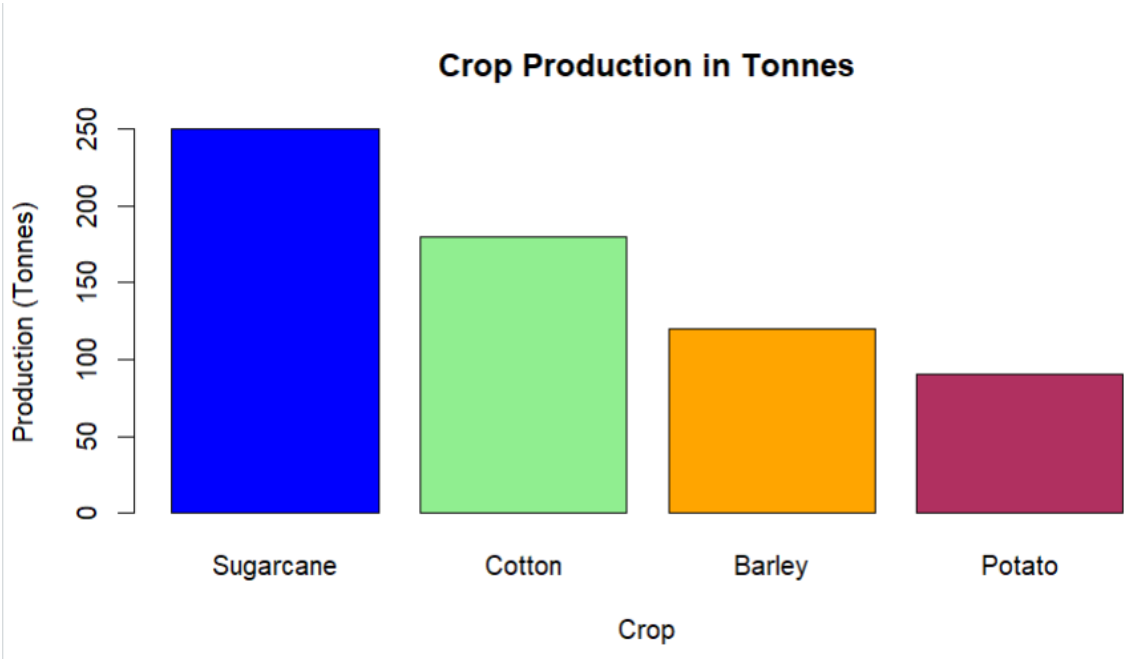

Grade Distribution
A
S
C
B


Number of Students per Grade
Number of Students
0
5
10
15
20
25
30
35
S
A
B
C
Grade


Employee Experience Distribution
1-3 years
4-6 years
10+ years
7-9 years


Number of Employees by Experience
Number of Employees
0
10
20
30
40
50
1-3 years
4-6 years
7-9 years
10+ years
Experience Level


Crop Production Distribution
Sugarcane
Cotton
Potato
Barley


Crop Production in Tonnes
Production (Tonnes)
0
50
100
150
200
250
Sugarcane
Cotton
Barley
Potato
Crop

```
> students <- data.frame(
+   Name =
+     c("KIRAN","DIVYA","ARUN","MEENA","RAHUL","SNEHA","VARUN","POOJA","KAVYA","MANOJ"),
+   Maths = c(82,76,91,88,73,80,86,79,95,92),
+   Science = c(85,92,78,74,89,87,72,90,94,91),
+   English = c(80,75,83,90,96,78,85,88,72,95)
+ )
> mean_marks <- sapply(students[ ,2:4], mean)
> median_marks <- sapply(students[ ,2:4], median)
> sd_marks <- sapply(students[ ,2:4], sd)
> students$Total <- rowSums(students[ ,2:4])
> top_student <- students[which.max(students$Total), ]
> avg_math <- mean(students$Maths)
> above_avg_math <- students[students$Maths > avg_math, ]
> mean_marks
  Maths Science English
   84.2    85.2    84.2
> median_marks
  Maths Science English
     84      88      84
> sd_marks
   Maths  Science  English
7.330302 7.814516 8.134973
> top_student
    Name Maths Science English Total
10 MANOJ    92      91      95   278
> above_avg_math
    Name Maths Science English Total
3   ARUN    91      78      83   252
4  MEENA    88      74      90   252
7  VARUN    86      72      85   243
9  KAVYA    95      94      72   261
10 MANOJ    92      91      95   278
> 9.2: Retail Sales Data Analysis
```

```
> sales <- data.frame(
+   Product = c("TV","Fridge","Microwave","Fan","AC","Mixer","Grinder"),
+   Day1 = c(18,30,20,12,25,15,28),
+   Day2 = c(16,25,60,14,40,35,22),
+   Day3 = c(28,48,52,35,55,38,30),
+   Day4 = c(38,50,45,13,15,28,34),
+   Day5 = c(45,38,30,20,50,40,26)
+ )
> sales_stats <- data.frame(
+   Product = sales$Product,
+   Mean = apply(sales[ ,2:6], 1, mean),
+   Max = apply(sales[ ,2:6], 1, max),
+   Min = apply(sales[ ,2:6], 1, min),
+   Variance = apply(sales[ ,2:6], 1, var)
+ )
> max_avg_product <- sales_stats[which.max(sales_stats$Mean), ]
> sales_stats
    Product Mean Max Min Variance
1        TV 29.0  45  16    157.0
2    Fridge 38.2  50  25    119.2
3 Microwave 41.4  60  20    264.8
4       Fan 18.8  35  12     91.7
5        AC 37.0  55  15    282.5
6     Mixer 31.2  40  15    102.7
7   Grinder 28.0  34  22     20.0
> max_avg_product
    Product Mean Max Min Variance
3 Microwave 41.4  60  20    264.8
> 9.3: Hospital Patient Age Analysis
```

```
> patients <- data.frame(
+   PatientID = 301:310,
+   Name = c("KIRAN","MEENA","ARUN","SURESH","RAVI","KAVYA","GANESH","LATHA","VIKRAM","ANIT
A"),
+   Age = c(42,36,55,30,62,47,53,38,44,49)
+ )
> mean_age <- mean(patients$Age)
> median_age <- median(patients$Age)
> range_age <- range(patients$Age)
> sd_age <- sd(patients$Age)
> older_patients <- patients[patients$Age > 50, ]
> mean_age
[1] 45.6
> median_age
[1] 45.5
> range_age
[1] 30 62
> sd_age
[1] 9.628661
> older_patients
  PatientID   Name Age
3       303   ARUN  55
5       305   RAVI  62
7       307 GANESH  53
> 9.4: Employee Salary Analysis
```

```
> employees <- data.frame(
+   EmpID = 201:210,
+   Name =
+     c("ARUN","KAVYA","MEENA","DIVYA","RAHUL","SURESH","ANITA","KIRAN","POOJA","MANOJ"),
+   Dept = c("HR","IT","Finance","HR","IT","Finance","HR","IT","Finance","HR"),
+   Salary = c(520000,720000,250000,340000,310000,760000,820000,400000,450000,360000)
+ )
> dept_stats <- aggregate(Salary ~ Dept, data=employees, function(x) c(Mean=mean(x),
+                                                                      Median=median(x)))
> dept_stats <- do.call(data.frame, dept_stats)
> overall_avg <- mean(employees$Salary)
> high_earners <- employees[employees$Salary > overall_avg, ]
> dept_stats
     Dept Salary.Mean Salary.Median
1 Finance    486666.7        450000
2      HR    510000.0        440000
3      IT    476666.7        400000
> high_earners
  EmpID   Name    Dept Salary
1   201   ARUN      HR 520000
2   202  KAVYA      IT 720000
6   206 SURESH Finance 760000
7   207  ANITA      HR 820000
> 9.5: Online Store Customer Ratings
```

```
> ratings <- data.frame(
+   ProductID = 11:20,
+   Product =
+     c("TV","Fridge","Microwave","Fan","AC","Mixer","Grinder","Iron","Cooler","Heater"),
+   Rating = c(8,6,5,7,4,6,5,9,8,7)
+ )
> mean_rating <- mean(ratings$Rating)
> median_rating <- median(ratings$Rating)
> var_rating <- var(ratings$Rating)
> getmode <- function(v) {
+   uniqv <- unique(v)
+   uniqv[which.max(tabulate(match(v, uniqv)))]
+ }
> mode_rating <- getmode(ratings$Rating)
> top_rated <- ratings[ratings$Rating > 5, ]
> mean_rating
[1] 6.5
> median_rating
[1] 6.5
> mode_rating
[1] 8
> var_rating
[1] 2.5
> top_rated
   ProductID Product Rating
1         11      TV      8
2         12  Fridge      6
4         14     Fan      7
6         16   Mixer      6
8         18    Iron      9
9         19  Cooler      8
10        20  Heater      7
```